

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

**Duration : 1 Hour
Total Marks:50**

Annexure A

Analytical Problem Solving

Code of Conduct:

I G. Pallavi (192311208) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Pallavi G



Annexure A

Student 6: GUDE PALLAVI | Register Number: 192311208

1. Solve the Missionaries and Cannibals problem: 3 missionaries and 3 cannibals must cross a river using a boat that can carry at most 2 people, such that cannibals never outnumber missionaries on either bank. Formulate the problem as a search problem (state, initial state, actions, transition model, goal test) and show the first three states explored using Breadth-First Search.
2. An e-commerce company wants to deploy a delivery drone to autonomously deliver packages within a city. Identify what type of problem-solving agent is suitable for this scenario, and write pseudocode outlining how the agent formulates and solves the delivery routing problem.
3. Given a graph with 8 nodes and a start node S and goal node G, trace Depth-Limited Search (DLS) with a depth limit of 3. Show which nodes are expanded and explain what happens if the goal lies beyond the depth limit.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

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Missionaries and Cannibals problem:-

a) Problem Formulation:-

State :

A State is represented as :

(M, C, B)

where :

- M = No. of missionaries on the left bank.
- C = Number of cannibals on the left bank.
- B = Boat position (left or right).

Example :-

- $(3, 3, L)$ = All missionaries, cannibals, and boat are on the left bank.

Initial State :-

$(3, 3, L)$

Actions:-

The boat can carry at most 2 people.

possible actions :

- Move 1 Missionary (1M)
- Move 2 Missionaries (2M)
- Move 1 cannibal (1C)
- Move 2 cannibals (2C)
- Move 1 missionary + 1 cannibal (1M1C)

Delivery Drone Problem-Solving Agent :-

Suitable Agent :-

A Goal-based problem-solving Agent is most suitable.

Reason:-

The drone has a clear goal :

→ Deliver package safely.

→ Find shortest route

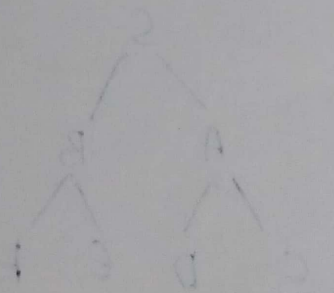
→ Avoid obstacles

→ Minimise delivery time

⇒ The agent searches for the best path before executing it.

Working Steps:-

1. Perceive delivery request.
2. Determine current location.
3. Identify destination
4. Build city map.
5. Search for shortest path.
6. Follow planned route.
7. Deliver package.
8. Return or wait for the next task.



Depth levels :

Depth 0 : S

Depth 1 : A B

Depth 2 : C D E F

Depth 3 : G

Suppose Goal = G

DLS Traversal

Start :- S

Expand :-
S
↓
A
↓
C
↓
G (Goal Found)

Expanded nodes : S A G C

⇒ If Goal Lies Beyond Depth 3

Example :-

S
↓
A
↓
C
↓
D
↓
G

⇒ Goal is at Depth 4 while the limit is 3.

Result :-

⇒ Goal is not found.

⇒ DLS returns Cutoff, indicating the search was limited by the depth bound rather than proving no solution exists.